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Cold treatment as a method of increasing the wear resistance of hydraulic breaker picks

Le Thanh Binh, V.I. Bolobov, G.A. Yusupov

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Abstract

It is shown that cold treatment of hydraulic breaker pick materials – steels 38XM, 40XH, 55C2, У8 and X12МФА (5-hour holding of specimens at −75 °C) – reduces the retained austenite content (by 44–88%) and increases the carbide content (by 15–98%) in hardened steel, which is accompanied by a substantial increase in the abrasive wear resistance of the materials (by 11–49%) when worn against granite, the main non-metallic raw material of Vietnam.

Keywords: *hydraulic breaker picks, carbon and alloy steels, cold treatment, increase in wear resistance.*

When active-action hydraulic breakers are operated on the hard and abrasive rocks typical of Vietnam's non-metallic raw-material deposits, the main cause of failure of the working tool – the picks – is their limiting wear (Fig. 1), caused by the insufficient wear resistance of the materials used – high-carbon and alloy steels.

[Figure 1 — see original PDF, page 2] ATLAS COPCO hydraulic breaker pick made of steel 40XH with a worn head part (left part of the pick) after 58 days of work breaking oversized granite blocks at a quarry in the Binh Dinh area (Vietnam)

It is known [1-3] that if the martensite finish point M_k of a steel lies below room temperature, then after quenching retained austenite remains in the structure of the material, and its amount increases with increasing content of carbon and alloying elements in the steel. The presence of retained austenite in the structure of the material of hardened parts adversely affects its physical and mechanical properties and manifests itself, first of all, in reduced hardness and, as a consequence, reduced wear resistance.

Cold treatment, which consists in cooling hardened steel below its martensite finish temperature M_k, promotes the transformation of retained austenite into martensite and the formation, during subsequent tempering, of a large number of very fine carbide particles, which also contributes to improving the service properties of the steel [8, 13]. For this reason, cold treatment is widely used as an additional operation after quenching to improve the wear resistance of tool steels [1, 4-7, 9-12, 14]. At the same time, no information on its use for improving the service properties of mining tool materials subjected to abrasive wear has been found in the literature. The present work compares the structure, mechanical properties ($\sigma_{0.2}$, σ_B, KCV, HRC) and abrasive wear resistance (R_w) of several steels most widely used as pick materials (38XM, 40XH, 55C2, У8 and X12МФА) after the heat treatment usually applied in practice (oil quenching from the required austenitizing temperature

plus low tempering at 200 °C) and after additional cold treatment. In the latter case, the specimens were placed, immediately after quenching and before tempering, in a cryostat with a cooling liquid at a temperature of −75 °C, where they were held for 5 hours.

Using an Axioplan-2 microscope with KS 300 3.0 software, the area fraction S_K of the steel microsection occupied by carbide grains was determined and taken as the percentage content of this phase. Then the total area fraction S_Σ of carbide and retained austenite grains was established, from which, by subtracting S_K, the content of the latter was calculated ($S_A = S_\Sigma - S_K$). Physical and mechanical characteristics were determined on standard five-fold specimens using a Zwick/Roell Z100 tensile testing machine, a Zwick/Roell RKP450 pendulum impact tester and a Zwick/Roell ZHU hardness tester. Wear resistance tests were carried out according to the method of the American standard ASTM G99 [5] (the “pin-on-disk” method) on cylindrical steel pin specimens (diameter 4 mm, height 14 mm), which, using a TE97 pin-on-disk machine, were worn under a constant load (P = 40 N) at a speed of 0.257 m/s against a granite disk, granite being the main non-metallic raw material of Vietnam. From the results of weighing the specimens before and after exposure to the abrasive, the wear resistance was calculated by the formula [5]:

$$R_w = \rho \cdot P \cdot L / m,\ N \cdot m/mm^3$$

where ρ is the density of the pin material, g/mm^3; m is the mass loss of the pin due to wear, g; L is the wear path (1000 m).

The results of the microstructural studies are presented in Figs. 2 and 3 and in Table 1.

[Figure 2 — see original PDF, page 3] Carbide content in steel X12МФА without (a) and with (b) cold treatment

The obtained values of the physical and mechanical characteristics and wear resistance of the steels, as the arithmetic mean of the results of 3 experiments, are presented in Table 2.

[Figure 3 — see original PDF, page 4] Total content of carbide and retained austenite in steel X12МФА without (a) and with (b) cold treatment

Table 1

Steel grade	Austenitizing temperature before quenching	% carbides		% retained austenite	
		without cold	with cold	without cold	with cold
38XM	950 °C	0.152÷0.182 0.168	0.281÷0.397 0.333 (+98.2%)	–	–
У8	800 °C	–	–	2.110÷2.314 2.212	1.258÷1.230 1.244 (−43.8%)
X12МФА	1000 °C	4.429÷5.847 5.616	6.095÷6.986 6.461 (+15.0%)	15.446	1.893 (−87.7%)

Note: the numerator of the fraction (upper line) is the limiting content of the analysed phase; the denominator (lower line) is the arithmetic mean of its 3 values.

Table 2

Steel grade	Treatment	Parameter values				
		Yield strength $\sigma_{0.2}$, MPa	Tensile strength σ_B, MPa	Impact toughness KCV, J/cm^2	Hardness, HRC	Wear resistance R_w, N·m/mm^3
38XM	Quenching from 870 °C + tempering	1478	1933	40.6	51.1	19075
	Quenching from 870 °C + cold + tempering	1529 (+3.5%)	1953 (+1.0%)	33.0 (−18.7%)	51.6 (+1.0%)	22763 (+19.3%)
40XH	Quenching from 820 °C + tempering	1759	2060	18.7	54.2	21889
	Quenching from 820 °C + cold + tempering	1719 (−2.3%)	2140 (+3.9%)	14.5 (−22.5%)	55.0 (+1.5%)	26240 (+19.9%)
55C2	Quenching from 870 °C + tempering	*)	*)	5.6	59.6	22542
	Quenching from 870 °C + cold + tempering	*)	*)	4.8 (−14.3%)	60.5 (+1.5%)	25220 (+11.9%)
У8	Quenching from 870 °C + tempering	*)	*)	5.1	60.4	13163
	Quenching from 870 °C + cold + tempering	*)	*)	3.9 (−23.5%)	60.9 (+0.8%)	14582 (+10.8%)
Х12МФА	Quenching from 870 °C + tempering	*)	*)	3.1	59.8	54740
	Quenching from 870 °C + cold + tempering	*)	*)	2.5 (−19.4%)	60.4 (+1.0%)	81722 (+49.3%)

Note: *) – brittle fracture of specimens over a wide range of stresses.

As follows from Figs. 2 and 3 and Tables 1 and 2, cold treatment, as in [1, 2, 3], leads to a significant decrease in the retained austenite content (by 44–88%) and an increase in the carbide content (by 15–98%) in hardened steel. This change in structure leads to a substantial increase in the wear resistance of the steels studied (by 11–49%) and is accompanied by a slight increase in the strength characteristics ($\sigma_{0.2}$, σ_B, HRC) of the tested materials, with a simultaneous decrease in impact toughness. (The $\sigma_{0.2}$ and σ_B values of steels 55C2, У8 and Х12МФА could not be determined

because of the large scatter in the experimental results caused by embrittlement of the specimen material as a result of heat treatment.)

From the research results presented, it can be concluded that cold treatment can be regarded as a promising additional heat-treatment operation for increasing the wear resistance of steels used to manufacture picks of hydraulic breakers working on hard, highly abrasive rocks. In doing so, the most optimal heat-treatment modes should be found at which the negative effect of cold treatment on the impact toughness of the steel will be minimal.

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